Cumulant Collision Code Verification: 10-Question Analysis

Cross-check of Gehrke (2024) Thesis §3.2 vs `cumulant_collision.h`

March 13, 2026

Abstract

This document provides a comprehensive technical analysis of the D3Q27 Cumulant Lattice Boltzmann Method (LBM) collision operator implementation against the theoretical framework presented in Gehrke (2024) §3.2. Each of the 10 questions addresses critical implementation details, comparing the thesis derivations with the actual code implementation in `cumulant_collision.h`. The analysis identifies one critical bug (Q2), several warning-level issues with parameterizations (Q1, Q6, Q9), and confirms correctness of other implementations (Q3–Q5, Q7–Q8). Specific code fixes and recommendations are provided for each issue.

Contents

1 Question 1: 4th-Order Diagonal Equilibria — Thesis Velocity Derivative Form vs Code A,B Product Form **2**
- 1.1 Thesis References . . . 2
- 1.2 Code Location . . . 2
- 1.3 Verdict . . . 2
- 1.4 Detailed Analysis . . . 2
  - 1.4.1 Thesis Form (Eq. 3.83–3.85) . . . 2
  - 1.4.2 Code Form (Lines 527–530) . . . 2
  - 1.4.3 Equivalence Analysis . . . 3
- 1.5 Recommended Fix . . . 3

2 Question 2: WP Mode Off-Diagonal 4th-Order Missing Low-Order Product Terms **3**
- 2.1 Thesis Reference . . . 3
- 2.2 Code Location . . . 3
- 2.3 Verdict . . . 4
- 2.4 Detailed Analysis . . . 4
  - 2.4.1 Cumulant-to-Central-Moment Relation . . . 4
  - 2.4.2 Current AO Mode Implementation (Lines 617–622) — CORRECT . . . 4
  - 2.4.3 Current WP Mode Implementation (Lines 613–615) — BUGGY . . . 4

2.4.4 Order-of-Magnitude Analysis . . . 5
2.5 Recommended Fix . . . 5

3 Question 3: Half-Force Correction Location 5
3.1 Thesis Reference . . . 5
3.2 Code Location . . . 6
3.3 Verdict . . . 6
3.4 Detailed Analysis . . . 6
3.4.1 Strang Splitting Approach . . . 6
3.4.2 Equivalence to Explicit Forcing . . . 6
3.4.3 Verification of Code Details . . . 7
3.4.4 GILBM vs Standard LBM . . . 7
3.5 Recommended Action . . . 7

4 Question 4: Velocity Derivative Equations 3.73–3.75 8
4.1 Thesis Reference . . . 8
4.2 Code Location . . . 8
4.3 Verdict . . . 8
4.4 Detailed Analysis . . . 8
4.4.1 Code Implementation . . . 8
4.4.2 Algebraic Verification . . . 8
4.4.3 Efficiency Gain . . . 9
4.5 Verdict Confirmation . . . 9

5 Question 5: Tilde-Omega Parameters (Eq. 3.111–3.114) 10
5.1 Thesis Reference . . . 10
5.2 Code Location . . . 10
5.3 Verdict . . . 10
5.4 Detailed Analysis . . . 10
5.4.1 Limiter Function Implementation . . . 10
5.4.2 Application to All Cumulant Pairs . . . 11
5.4.3 Cumulant-to-Rate Correspondence . . . 11
5.4.4 Lambda Parameter . . . 11
5.5 Verdict Confirmation . . . 12

6 Question 6: 4th-Order Diagonal — Velocity Derivative Terms vs ω_6 Relaxation 12
6.1 Thesis Reference . . . 12
6.2 Code Location . . . 12
6.3 Verdict . . . 12
6.4 Detailed Analysis . . . 12
6.4.1 Thesis Form (Eq. 3.83 Expanded) . . . 12
6.4.2 Standard Collision Operator Form . . . 13
6.4.3 Equivalence at $\omega_6 = \omega_7 = 1$ . . . 13
6.4.4 Divergence for $\omega_6 \neq 1$ . . . 13

6.5 Current Parameter Status . . . 14
6.6 Recommended Fix . . . 14

7 Question 7: Keep ω_8 Explicit in Eq. 3.86–3.88 14
7.1 Thesis References . . . 14
7.2 Code Location . . . 14
7.3 Verdict . . . 14
7.4 Detailed Analysis . . . 15
7.4.1 Eq. 3.89 Velocity Derivative Relations — THESIS TYPO IDENTIFIED 15
7.4.2 Eq. 3.86–3.88 General Form (Keeping ω_8) . . . 15
7.4.3 Current Code (Simplified with $\omega_8 = 1$) . . . 15
7.4.4 Independence of ω_8 and Velocity Derivatives . . . 16
7.5 Recommended Implementation . . . 16
7.6 Additional Requirements . . . 16

8 Question 8: Eq. 3.108–3.110 — Automatic Behavior 17
8.1 Thesis Reference . . . 17
8.2 Code Location . . . 17
8.3 Verdict . . . 17
8.4 Detailed Analysis . . . 17
8.4.1 Analytical Behavior . . . 17
8.4.2 Automatic in Code . . . 18
8.4.3 Lambda Limiter Counteracts Unconditional Conservation . . . 18
8.4.4 Safety Clamp . . . 18
8.5 Verdict Confirmation . . . 18

9 Question 9: ω_6, ω_7 General Form Claim 19
9.1 Thesis References . . . 19
9.2 Code Location . . . 19
9.3 Verdict . . . 19
9.4 Detailed Analysis . . . 19
9.4.1 Thesis Structure . . . 19
9.4.2 Code Deviatoric/Trace Decomposition . . . 19
9.4.3 Equivalence Constraint . . . 20
9.4.4 Parameter Status . . . 20
9.4.5 Thesis Constraint . . . 20
9.5 Recommended Fix . . . 20

10 Question 10: Generalize Eq. 3.86–3.88 with ω_8 Parameter 21
10.1 Thesis References . . . 21
10.2 Code Location . . . 21
10.3 Verdict . . . 21
10.4 Detailed Analysis . . . 21
10.4.1 Step 1: Velocity Derivatives from Eq. 3.89 (CORRECTED) . . . 21
10.4.2 Step 2: General Eq. 3.86–3.88 with ω_8 . . . 22

10.4.3 Step 3: Back-Conversion with Product Terms (from Q2) 22
10.4.4 Verification: Simplified Case $\omega_8 = 1$ 23
10.4.5 Implementation Checklist 23
10.5 Recommended Complete Code Section 24

1 Question 1: 4th-Order Diagonal Equilibria — Thesis Velocity Derivative Form vs Code A,B Product Form

1.1 Thesis References

- **General Form**: Eq. 3.83–3.85 (pages 45–46)
- **Simplified Form**: Eq. 3.116–3.118 (page 47) with constraint $\omega_6 = \omega_7 = 1$

1.2 Code Location

`cumulant_collision.h`, lines 527–530 (4th-order diagonal equilibrium computation)

1.3 Verdict

WARNING/LIMITATION

Different parameterization, equivalent ONLY when $\omega_6 = \omega_7 = 1$

1.4 Detailed Analysis

1.4.1 Thesis Form (Eq. 3.83–3.85)

The thesis employs a velocity-derivative formulation for the 4th-order diagonal equilibria:

$$\begin{aligned}\kappa^*_{220} = \frac{2}{3}\left(\frac{1}{\omega_1} - \frac{1}{2}\right) A\rho(\partial_x u - 2\partial_y v + \partial_z w) \\ + (1-\omega_6)(C_{220} - 2C_{202} + C_{022}) + \frac{1}{3}\big[\cdots\big]_{\text{trace mode}}\end{aligned} \tag{1}$$

The structure decomposes the collision operator into:

1. Velocity derivative equilibrium term (explicit $\partial_x u$, $\partial_y v$, $\partial_z w$)
2. Deviatoric mode relaxation: $(1-\omega_6)\cdot C_{\text{dev}}$
3. Trace mode relaxation: $(1-\omega_7)\cdot C_{\text{trace}}$

1.4.2 Code Form (Lines 527–530)

The implementation uses a product form from [3]:

```
CUMcca_eq = (A*(Sxx*Syy + Dxy_sq) + B*(Sxx*Syy - Dxy_sq)) / rho;
```

Listing 1: 4th-order diagonal equilibrium (cumulant_collision.h:527–530)

where $S_{xx} = 1/3 +$ shear terms, $D_{xy} =$ post-relaxation shear products, and A, B are constants.

1.4.3 Equivalence Analysis

The relationship between the two forms can be established through the standard LBM collision operator:

$$C^* = (1-\omega)C + \omega C^{\text{eq}} \tag{2}$$

For the code's A,B product form to equal the thesis's velocity-derivative form:

$$\omega_6 \cdot C^{\text{eq}}_{\text{code}} = \text{velocity_derivative_equilibrium}_{\text{thesis}} \tag{3}$$

At equilibrium ($S_{xx} = S_{yy} = 1/3$, $D_{xy} = 0$):

$$C^{\text{eq}}_{\text{code}} = \frac{A+B}{9\rho} \neq 0 \tag{4}$$

This non-zero offset must be absorbed into the initial condition via `cumulant_equilibrium_host.h`.

Key constraint: Equivalence requires $\omega_6 = \omega_7 = 1$. For $\omega_6 \neq 1$, the code's form and thesis form diverge because the thesis's velocity derivative terms are *not* divided by ω_6.

1.5 Recommended Fix

Keep the [GR22] A,B formulation but update documentation:

```
// IMPORTANT: A,B product form equivalent to Eq. 3.83-3.85 ONLY at omega6=omega7=1
// For general omega6 != 1 or omega7 != 1, must revert to explicit
// velocity derivative equilibrium form from thesis Eq. 3.83-3.85.
// Current constraint: omega6 = omega7 = 1 (set at initialization)
CUMcca_eq = (A*(Sxx*Syy + Dxy_sq) + B*(Sxx*Syy - Dxy_sq)) / rho;
```

Listing 2: Updated comment for 4th-order diagonal equilibrium

—

2 Question 2: WP Mode Off-Diagonal 4th-Order Missing Low-Order Product Terms

2.1 Thesis Reference

Eq. J.16 (Geier 2015 Appendix J): Cumulant-to-central-moment back-conversion requires adding products of lower-order cumulants

2.2 Code Location

- **AO mode (correct)**: `cumulant_collision.h`, lines 617–622
- **WP mode (BUGGY)**: `cumulant_collision.h`, lines 613–615

2.3 Verdict

BUG

CRITICAL BUG: Missing cumulant-to-central-moment back-conversion in WP mode

2.4 Detailed Analysis

2.4.1 Cumulant-to-Central-Moment Relation

The relationship between 4th-order off-diagonal cumulants and central moments (Eq. J.16, Geier 2015):

$$\mu_{211} = \kappa_{211} + (\kappa_{200} \cdot \kappa_{011} + 2\kappa_{110} \cdot \kappa_{101})/\rho \tag{5}$$

In code variables (well-conditioned form):

$$m[I_{\text{cbb}}] = C^*_{211} + \frac{(m[I_{\text{caa}}] + 1/3) \cdot m[I_{\text{abb}}] + 2 \cdot m[I_{\text{bba}}] \cdot m[I_{\text{bab}}]}{\rho} \tag{6}$$

2.4.2 Current AO Mode Implementation (Lines 617–622) — CORRECT

```
// AO mode: back-convert cumulant to central moment (CORRECT)
m[I_cbb] = CUMcbb + ((m[I_caa]+1.0/3.0)*m[I_abb] +
                     2.0*m[I_bba]*m[I_bab]) * inv_rho;
m[I_bcb] = CUMbcb + ((m[I_aca]+1.0/3.0)*m[I_bab] +
                     2.0*m[I_bba]*m[I_abb]) * inv_rho;
m[I_bbc] = CUMbbc + ((m[I_aac]+1.0/3.0)*m[I_bba] +
                     2.0*m[I_bab]*m[I_abb]) * inv_rho;
```

Listing 3: AO mode: correct back-conversion (cumulant_collision.h:617–622)

2.4.3 Current WP Mode Implementation (Lines 613–615) — BUGGY

```
// WP mode: BUGGY     direct assignment without cumulant->central
   moment conversion
m[I_cbb] = wp_C211_star;  // MISSING product terms!
m[I_bcb] = wp_C121_star;  // MISSING product terms!
m[I_bbc] = wp_C112_star;  // MISSING product terms!
```

Listing 4: WP mode: MISSING back-conversion (cumulant_collision.h:613–615)

2.4.4 Order-of-Magnitude Analysis

The missing product terms are NOT negligible:

1. **Magnitude of WP correction**:

$$wp_C211_star \approx (\omega/2 - 1)\omega BC_{011} = O(Ma) \tag{7}$$

   where Ma is the Mach number.

2. **Magnitude of product terms**:

$$(m[I_{\text{caa}}] + 1/3) \approx 1/3 + O(Ma^2) \tag{8}$$
$$m[I_{\text{abb}}] \approx O(Ma) \quad \text{(post-relaxation, reduced by } (1-\omega_1)) \tag{9}$$
$$\text{Product} \approx O(Ma)/3 \quad \text{per component} \tag{10}$$

3. **Comparison**: Both the WP correction and the missing product terms are $O(Ma)$, making them *equally significant*.

4. **Near walls**: Large velocity gradients amplify the 2nd-order cumulants, making the products even more important.

2.5 Recommended Fix

Replace WP mode lines 613–615 with:

```
// WP mode: FIXED     add cumulant-to-central moment back-conversion
m[I_cbb] = wp_C211_star + ((m[I_caa]+1.0/3.0)*m[I_abb] +
                           2.0*m[I_bba]*m[I_bab]) * inv_rho;
m[I_bcb] = wp_C121_star + ((m[I_aca]+1.0/3.0)*m[I_bab] +
                           2.0*m[I_bba]*m[I_abb]) * inv_rho;
m[I_bbc] = wp_C112_star + ((m[I_aac]+1.0/3.0)*m[I_bba] +
                           2.0*m[I_bab]*m[I_abb]) * inv_rho;
```

Listing 5: WP mode fix: add back-conversion products

—

3 Question 3: Half-Force Correction Location

3.1 Thesis Reference

Section 3.2.1, page 43: "Additional body forces are considered by augmenting the velocity preliminary to the forward transformation $f_\zeta \to \kappa_{\alpha\beta\gamma} = \kappa_{\alpha\beta\gamma}(u + F/(2\rho))$. Moreover, the three first-order central moments ($\kappa_{\bar{o}=1}$) have to change sign prior to the back transformation in this case."

3.2 Code Location

- Lines 234–236: Velocity augmentation
- Line 252: Forward chimera with augmented velocity
- Lines 684–686: Sign flip of 1st-order cumulants
- Lines 714–729: Guo source term (DISABLED)

3.3 Verdict

CORRECT

Correct implementation using Strang splitting

3.4 Detailed Analysis

3.4.1 Strang Splitting Approach

The code implements a Strang splitting to incorporate body forces without an explicit source term:

1. **Step 1 (Half-force velocity augmentation)**: Line 234–236

$$\tilde{u} = u + \frac{F\Delta t}{2\rho} \tag{11}$$

2. **Step 2 (Forward chimera with $\tilde{u}$)**: Line 252

$$\kappa_{\alpha\beta\gamma} = \kappa_{\alpha\beta\gamma}(\tilde{u}) \tag{12}$$

3. **Step 3 (Normal collision in cumulant space)**: Lines 254–689

$$\kappa^*_{\alpha\beta\gamma} = (1-\omega_i)\kappa_{\alpha\beta\gamma} + \omega_i \kappa^{\text{eq}}_{\alpha\beta\gamma} \tag{13}$$

4. **Step 4 (Sign flip of 1st-order moments)**: Lines 684–686

$$\kappa^*_{100} \to -\kappa^*_{100}, \quad \kappa^*_{010} \to -\kappa^*_{010}, \quad \kappa^*_{001} \to -\kappa^*_{001} \tag{14}$$

5. **Step 5 (Backward chimera)**: Line 714

$$f_i^{n+1} = f_i^* \quad \text{(in original velocity frame)} \tag{15}$$

3.4.2 Equivalence to Explicit Forcing

This Strang splitting encodes the second half of the force correction:

$$\text{(Half-force forward)} + \text{(Sign flip)} = \text{(Explicit Guo source term removed)} \tag{16}$$

The thesis notes: "This reduces the number of operations by a factor of three" (page 43).

3.4.3 Verification of Code Details

Line 234–236 (Velocity augmentation):

```
// Augment velocity with half-force (for GILBM, multiply by  t )
u_aug_x += 0.5 * F_x * dt_over_rho;
u_aug_y += 0.5 * F_y * dt_over_rho;
```

Listing 6: Half-force velocity augmentation (cumulant_collision.h:234–236)

Line 684–686 (Sign flip):

```
// Sign flip of 1st-order cumulants before backward chimera
kappa100 = -kappa100;
kappa010 = -kappa010;
kappa001 = -kappa001;
```

Listing 7: Sign flip of 1st-order moments (cumulant_collision.h:684–686)

Line 714–729 (Guo source disabled):

```
#if 0  // DISABLED: Strang splitting replaces explicit forcing
// Guo forcing source term (NOT USED in this implementation)
// fpost_i += Guo_weight[i] * (c_x * F_x + c_y * F_y + c_z * F_z) /
   cs2
#endif
```

Listing 8: Guo source term disabled (cumulant_collision.h:714–729)

3.4.4 GILBM vs Standard LBM

The code correctly includes the Δt factor in the velocity augmentation:

$$\tilde{u} = u + \frac{F\Delta t}{2\rho} \tag{17}$$

Standard LBM (with lattice units $\Delta t = 1$) would have:

$$\tilde{u} = u + \frac{F}{2\rho} \tag{18}$$

3.5 Recommended Action

No code changes needed. Add clarifying comment:

```
// Strang splitting for body forces (thesis Section 3.2.1)
// Instead of explicit Guo source term, we use:
// 1. Half-force velocity augmentation in forward chimera
// 2. Sign flip of 1st-order cumulants before backward chimera
// This encodes the complete force correction while reducing
   operations by 3x.
```

Listing 9: Suggested documentation comment

—

4 Question 4: Velocity Derivative Equations 3.73–3.75

4.1 Thesis Reference

- **Eq. 3.73**: $\partial_x u = [\omega_1(-2C_{200} + C_{020} + C_{002}) - \omega_2(C_{200} + C_{020} + C_{002} - \kappa_{000})]/(2\rho)$
- **Eq. 3.74**: $\partial_y v = [\omega_1(C_{200} - 2C_{020} + C_{002}) - \omega_2(C_{200} + C_{020} + C_{002} - \kappa_{000})]/(2\rho)$
- **Eq. 3.75**: $\partial_z w = [\omega_1(C_{200} + C_{020} - 2C_{002}) - \omega_2(C_{200} + C_{020} + C_{002} - \kappa_{000})]/(2\rho)$

4.2 Code Location

`cumulant_collision.h`, lines 432–435

4.3 Verdict

CORRECT

Correct — code uses efficient incremental form

4.4 Detailed Analysis

4.4.1 Code Implementation

```
double Dxux = -0.5*omega*inv_rho*(2*m[I_caa]-m[I_aca]-m[I_aac])
              - 0.5*omega2*inv_rho*(mxxPyyPzz-drho);
double Dyuy = Dxux + 1.5*omega*inv_rho*(m[I_caa]-m[I_aca]);
double Dzuz = Dxux + 1.5*omega*inv_rho*(m[I_caa]-m[I_aac]);
```

Listing 10: Velocity derivative computation (cumulant_collision.h:432–435)

4.4.2 Algebraic Verification

Expansion of Dxux:

$$\begin{aligned}
\text{Dxux} &= -\frac{\omega_1}{2\rho}(2C_{200} - C_{020} - C_{002}) - \frac{\omega_2}{2\rho}(C_{200} + C_{020} + C_{002} - \kappa_{000}) && (19)\\
&= \frac{\omega_1}{2\rho}(-2C_{200} + C_{020} + C_{002}) - \frac{\omega_2}{2\rho}(\text{trace} - \kappa_{000}) && (20)\\
&= \partial_x u \quad \checkmark \quad (\text{Eq. } 3.73) && (21)
\end{aligned}$$

Expansion of Dyuy:

$$\begin{aligned}
\text{Dyuy} &= \text{Dxux} + \frac{3\omega_1}{2\rho}(C_{200} - C_{020}) && (22)\\
&= \frac{\omega_1}{2\rho}(-2C_{200} + C_{020} + C_{002}) + \frac{3\omega_1}{2\rho}(C_{200} - C_{020}) && (23)\\
&\quad - \frac{\omega_2}{2\rho}(\text{trace} - \kappa_{000}) && (24)\\
&= \frac{\omega_1}{2\rho}(C_{200} - 2C_{020} + C_{002}) - \frac{\omega_2}{2\rho}(\text{trace} - \kappa_{000}) && (25)\\
&= \partial_y v \quad \checkmark \quad (\text{Eq. } 3.74) && (26)
\end{aligned}$$

Expansion of Dzuz:

$$\begin{aligned}
\text{Dzuz} &= \text{Dxux} + \frac{3\omega_1}{2\rho}(C_{200} - C_{002}) && (27)\\
&= \frac{\omega_1}{2\rho}(-2C_{200} + C_{020} + C_{002}) + \frac{3\omega_1}{2\rho}(C_{200} - C_{002}) && (28)\\
&\quad - \frac{\omega_2}{2\rho}(\text{trace} - \kappa_{000}) && (29)\\
&= \frac{\omega_1}{2\rho}(C_{200} + C_{020} - 2C_{002}) - \frac{\omega_2}{2\rho}(\text{trace} - \kappa_{000}) && (30)\\
&= \partial_z w \quad \checkmark \quad (\text{Eq. } 3.75) && (31)
\end{aligned}$$

4.4.3 Efficiency Gain

The incremental form saves computation:

- Direct evaluation: 9 multiplications (3 per equation)
- Incremental form: 4 multiplications total + 2 additions
- **Savings**: ≈ 5 operations ($\sim 56\%$ reduction)

4.5 Verdict Confirmation

No changes needed. The code correctly implements Eq. 3.73–3.75 using an efficient incremental computation.

—

5 Question 5: Tilde-Omega Parameters (Eq. 3.111–3.114)

5.1 Thesis Reference

Eq. 3.111–3.114 (pages 46–47): Limiter functions for $\omega_3, \omega_4, \omega_5$

$$\tilde{\omega}_{3,1} = \omega_{3,1} + (1 - \omega_{3,1}) \frac{|C_{120} + C_{102}|}{\rho\lambda + |C_{120} + C_{102}|} \tag{32}$$

$$\tilde{\omega}_{4,1} = \omega_{4,1} + (1 - \omega_{4,1}) \frac{|C_{120} - C_{102}|}{\rho\lambda + |C_{120} - C_{102}|} \tag{33}$$

$$\tilde{\omega}_{3,2} = \omega_{3,2} + (1 - \omega_{3,2}) \frac{|C_{210} + C_{012}|}{\rho\lambda + |C_{210} + C_{012}|} \tag{34}$$

$$\tilde{\omega}_{4,2} = \omega_{4,2} + (1 - \omega_{4,2}) \frac{|C_{210} - C_{012}|}{\rho\lambda + |C_{210} - C_{012}|} \tag{35}$$

$$\tilde{\omega}_{3,3} = \omega_{3,3} + (1 - \omega_{3,3}) \frac{|C_{201} + C_{021}|}{\rho\lambda + |C_{201} + C_{021}|} \tag{36}$$

$$\tilde{\omega}_{4,3} = \omega_{4,3} + (1 - \omega_{4,3}) \frac{|C_{201} - C_{021}|}{\rho\lambda + |C_{201} - C_{021}|} \tag{37}$$

$$\tilde{\omega}_5 = \omega_5 + (1 - \omega_5) \frac{|C_{111}|}{\rho\lambda + |C_{111}|} \tag{38}$$

with $\lambda = 10^{-2}$ (Eq. 3.115).

5.2 Code Location

- Lines 393–409: Limiter function `_cum_wp_limit()`
- Lines 468–476: Application of $\tilde{\omega}$ to all 7 symmetric/antisymmetric pairs

5.3 Verdict

CORRECT

Correct — all 7 cumulant pairs matched to correct tilde-omega rates

5.4 Detailed Analysis

5.4.1 Limiter Function Implementation

```
static inline double _cum_wp_limit(double omega_base, double C_mag,
                                   double rho, double lambda) {
    // Eq. 3.111-3.114: tilde_omega = omega + (1-omega)*|C|/(rho*
       lambda + |C|)
```

```
    double abs_C = fabs(C_mag);
    double denominator = rho * lambda + abs_C;
    return omega_base + (1.0 - omega_base) * abs_C / denominator;
}
```

Listing 11: Limiter function _cum_wp_limit (cumulant_collision.h:393–409)

This directly implements the Eq. 3.111–3.114 formula:

$$\tilde{\omega} = \omega_{\text{base}} + (1 - \omega_{\text{base}}) \frac{|C_{\alpha\beta\gamma}|}{\rho\lambda + |C_{\alpha\beta\gamma}|} \tag{39}$$

5.4.2 Application to All Cumulant Pairs

```
omega3_1 = _cum_wp_limit(omega3_base, C_120 + C_102, rho, lambda);
omega4_1 = _cum_wp_limit(omega4_base, C_120 - C_102, rho, lambda);
omega3_2 = _cum_wp_limit(omega3_base, C_210 + C_012, rho, lambda);
omega4_2 = _cum_wp_limit(omega4_base, C_210 - C_012, rho, lambda);
omega3_3 = _cum_wp_limit(omega3_base, C_201 + C_021, rho, lambda);
omega4_3 = _cum_wp_limit(omega4_base, C_201 - C_021, rho, lambda);
omega5_lim = _cum_wp_limit(omega5_base, C_111, rho, lambda);
```

Listing 12: Tilde-omega computation for all pairs (cumulant_collision.h:393–409)

5.4.3 Cumulant-to-Rate Correspondence

Cumulant Pair	Code Variable	Tilde-Omega	Thesis Equation
$C_{120} + C_{102}$	`mxyyPxzz`	`omega3_1`	Eq. 3.76–3.77
$C_{120} - C_{102}$	`mxyyMxzz`	`omega4_1`	Eq. 3.76–3.77
$C_{210} + C_{012}$	`mxxyPyzz`	`omega3_2`	Eq. 3.78–3.79
$C_{210} - C_{012}$	`mxxyMyzz`	`omega4_2`	Eq. 3.78–3.79
$C_{201} + C_{021}$	`mxxzPyyz`	`omega3_3`	Eq. 3.80–3.81
$C_{201} - C_{021}$	`mxxzMyyz`	`omega4_3`	Eq. 3.80–3.81
C_{111}	`m[I_bbb]`	`omega5_lim`	Eq. 3.82

Table 1: Cumulant pairs and their corresponding tilde-omega relaxation rates

Verification of Eq. 3.76–3.77:

$$m^*[I_{\text{mxyyPxzz}}] = (1 - \tilde{\omega}_{3,1}) m[I_{\text{mxyyPxzz}}] \tag{40}$$

Similarly for all other pairs. Each pair is relaxed with its corresponding $\tilde{\omega}$ computed via the limiter function.

5.4.4 Lambda Parameter

The code sets $\lambda =$ `CUM_LAMBDA` $= 10^{-2}$ (line 78 in variables.h), matching Eq. 3.115.

5.5 Verdict Confirmation

All 7 cumulant pairs are correctly identified, and each is relaxed with the appropriate tilde-omega rate computed via Eq. 3.111–3.114. No changes needed.

—

6 Question 6: 4th-Order Diagonal — Velocity Derivative Terms vs ω_6 Relaxation

6.1 Thesis Reference

Eq. 3.83 (page 45): Explicit structure with ω_6 appearing as $(1-\omega_6)$ multiplying deviatoric modes

6.2 Code Location

Lines 527–530 (equilibrium computation) and lines 355–356 (ω_6, ω_7 parameter storage)

6.3 Verdict

WARNING/LIMITATION

Two DIFFERENT formulations — equivalent ONLY at $\omega_6 = \omega_7 = 1$

6.4 Detailed Analysis

6.4.1 Thesis Form (Eq. 3.83 Expanded)

The thesis gives the 4th-order diagonal equilibrium with explicit ω_6, ω_7 dependence:

$$\kappa^*_{220} = \frac{2}{3}\left(\frac{1}{\omega_1} - \frac{1}{2}\right) A\rho(\partial_x u - 2\partial_y v + \partial_z w) \tag{41}$$
$$+ (1-\omega_6)(C_{220} - 2C_{202} + C_{022})$$
$$+ \frac{2}{3}\left(\frac{1}{\omega_1} - \frac{1}{2}\right) A\rho(\partial_x u + \partial_y v - 2\partial_z w) \tag{42}$$
$$+ (1-\omega_6)(C_{220} + C_{202} - 2C_{022})$$
$$- \frac{4}{3}\left(\frac{1}{\omega_1} - \frac{1}{2}\right) A\rho(\partial_x u + \partial_y v + \partial_z w) \tag{43}$$
$$+ (1-\omega_7)(C_{220} + C_{202} + C_{022})/3$$

This structure decomposes into:

1. **Velocity derivative equilibrium**: Explicitly written functions of $\partial_x u, \partial_y v, \partial_z w$

2. **Deviatoric relaxation**: $(1 - \omega_6) \cdot C_{\text{dev}}$

3. **Trace relaxation**: $(1 - \omega_7) \cdot C_{\text{trace}}$

6.4.2 Standard Collision Operator Form

In the standard LBM collision operator:

$$C_i^* = (1 - \omega_i)C_i + \omega_i C_i^{\text{eq}} \tag{44}$$

For the thesis form to match the code, we need:

$$\omega_6 \cdot C_{\text{code}}^{\text{eq}} = \text{velocity_derivative_equilibrium}_{\text{thesis}} \tag{45}$$

6.4.3 Equivalence at $\omega_6 = \omega_7 = 1$

When $\omega_6 = \omega_7 = 1$, the thesis form simplifies to:

$$\kappa_{220}^* = \text{velocity_derivative_equilibrium} \tag{46}$$

The code's A,B product form:

$$C_{\text{code}}^{\text{eq}} = \frac{A(S_{xx}S_{yy} + D_{xy}^2) + B(S_{xx}S_{yy} - D_{xy}^2)}{\rho} \tag{47}$$

When substituted back into the collision operator with $\omega_6 = 1$:

$$C_{\text{code}}^* = (1 - 1)C + 1 \cdot C^{\text{eq}} = C^{\text{eq}} \tag{48}$$

which matches the thesis velocity derivative form (since the deviatoric and trace relaxation terms vanish with $(1 - 1) = 0$).

6.4.4 Divergence for $\omega_6 \neq 1$

For $\omega_6 \neq 1$, the code's form becomes:

$$C^* = (1 - \omega_6)C + \omega_6 \frac{A(S_{xx}S_{yy} + D_{xy}^2) + B(S_{xx}S_{yy} - D_{xy}^2)}{\rho} \tag{49}$$

To match the thesis, we would need:

$$\omega_6 C_{\text{code}}^{\text{eq}} = \text{velocity_derivative_equilibrium}_{\text{thesis}} \tag{50}$$

which requires dividing the equilibrium by ω_6. The code does *not* do this.

Conclusion: The code's A,B form is ONLY equivalent to the thesis's velocity derivative form when $\omega_6 = \omega_7 = 1$.

6.5 Current Parameter Status

Lines 355–356:

```
const double omega6 = CUM_OMEGA6;  // Default 1.0
const double omega7 = CUM_OMEGA7;  // Default 1.0
```

Listing 13: Parameter storage (cumulant_collision.h:355–356)

Currently hardcoded to 1.0, which maintains equivalence with the thesis.

6.6 Recommended Fix

Update documentation to clearly state the constraint:

```
// 4th-order diagonal equilibrium (deviatoric/trace decomposition)
// NOTE: A,B product form is equivalent to thesis Eq. 3.83-3.85
//       ONLY when omega6 = omega7 = 1 (currently enforced).
// For general omega6 != 1 or omega7 != 1, must revert to explicit
// velocity derivative equilibrium form from thesis Eq. 3.83-3.85
// with division by omega6 applied.
const double omega6 = CUM_OMEGA6;  // Constraint: must be 1.0 for A,
   B form
const double omega7 = CUM_OMEGA7;  // Constraint: must be 1.0 for A,
   B form
```

Listing 14: Revised comment for 4th-order diagonals

—

7 Question 7: Keep ω_8 Explicit in Eq. 3.86–3.88

7.1 Thesis References

- **Page 47, §3.2.3**: "The remaining higher-order ($\bar{o} > 3$) relaxation rates are set to unity, i.e., $\omega_{C>5} = 1$"
- **Eq. 3.86–3.88**: Off-diagonal 4th-order collision formulas with explicit ω_8 structure
- **Eq. 3.89**: Velocity derivative approximation (with TYPO in subscripts)

7.2 Code Location

Lines 566–571 (current simplified form with $\omega_8 = 1$ substituted)

7.3 Verdict

WARNING/LIMITATION

Valid simplification per thesis, but should keep ω_8 as explicit parameter

7.4 Detailed Analysis

7.4.1 Eq. 3.89 Velocity Derivative Relations — THESIS TYPO IDENTIFIED

The thesis (Eq. 3.89) provides Chapman–Enskog analysis results relating velocity derivatives to 2nd-order cumulants. **However, the subscripts are SWAPPED in the first and third relations.**

Equation	Thesis (TYPO)	Correct
First relation	$\partial_y w + \partial_z v = -3\omega_1/\rho \cdot C_{110}$	$\partial_y w + \partial_z v = -3\omega_1/\rho \cdot C_{011}$
Second relation	$\partial_x w + \partial_z u = -3\omega_1/\rho \cdot C_{101}$	$\partial_x w + \partial_z u = -3\omega_1/\rho \cdot C_{101}$
Third relation	$\partial_x v + \partial_y u = -3\omega_1/\rho \cdot C_{011}$	$\partial_x v + \partial_y u = -3\omega_1/\rho \cdot C_{110}$

Table 2: Thesis Eq. 3.89 with subscript corrections

Physical reasoning for the correction:

$$C_{011} = \langle (c_y - u_y)(c_z - u_z) \rangle_\rho \quad \text{(yz shear stress)} \quad \Rightarrow \quad \partial_y w + \partial_z v \tag{51}$$
$$C_{101} = \langle (c_x - u_x)(c_z - u_z) \rangle_\rho \quad \text{(xz shear stress)} \quad \Rightarrow \quad \partial_x w + \partial_z u \tag{52}$$
$$C_{110} = \langle (c_x - u_x)(c_y - u_y) \rangle_\rho \quad \text{(xy shear stress)} \quad \Rightarrow \quad \partial_x v + \partial_y u \tag{53}$$

7.4.2 Eq. 3.86–3.88 General Form (Keeping ω_8)

The general collision formula for off-diagonal 4th-order cumulants:

$$\kappa^*_{211} = -\frac{1}{3}(\omega_1/2 - 1)\omega_8 B\rho(\partial_y w + \partial_z v) + (1 - \omega_8)\kappa_{211} \tag{54}$$
$$\kappa^*_{121} = -\frac{1}{3}(\omega_1/2 - 1)\omega_8 B\rho(\partial_x w + \partial_z u) + (1 - \omega_8)\kappa_{121} \tag{55}$$
$$\kappa^*_{112} = -\frac{1}{3}(\omega_1/2 - 1)\omega_8 B\rho(\partial_x v + \partial_y u) + (1 - \omega_8)\kappa_{112} \tag{56}$$

Substituting the Eq. 3.89 relations (CORRECTED):

$$\kappa^*_{211} = ((\omega_1/2 - 1)\omega_1\omega_8 B C_{011}) + (1 - \omega_8)\kappa_{211} \tag{57}$$
$$\kappa^*_{121} = ((\omega_1/2 - 1)\omega_1\omega_8 B C_{101}) + (1 - \omega_8)\kappa_{121} \tag{58}$$
$$\kappa^*_{112} = ((\omega_1/2 - 1)\omega_1\omega_8 B C_{110}) + (1 - \omega_8)\kappa_{112} \tag{59}$$

7.4.3 Current Code (Simplified with $\omega_8 = 1$)

```
const double wp_offdiag_coeff = (omega*0.5 - 1.0) * omega * coeff_B;
wp_C211_star = wp_offdiag_coeff * saved_C011;
wp_C121_star = wp_offdiag_coeff * saved_C101;
wp_C112_star = wp_offdiag_coeff * saved_C110;
```

Listing 15: Current form with omega8

This matches the simplified case when $\omega_8 = 1$.

7.4.4 Independence of ω_8 and Velocity Derivatives

Key insight: The Eq. 3.89 velocity derivative relations come from Chapman–Enskog analysis of the *2nd-order collision* (controlled by ω_1), not from ω_8.

Therefore, Eq. 3.89 is *independent of* ω_8 and can be substituted into Eq. 3.86–3.88 regardless of the value of ω_8.

7.5 Recommended Implementation

Generalize to support arbitrary ω_8:

```
// Eq. 3.89 (CORRECTED for thesis typo in subscripts):
//   _yw   +   _zv   = -3     /      C_011  [thesis prints C_110 TYPO]
//   _xw   +   _zu   = -3     /      C_101  [correct]
//   _xv   +   _yu   = -3     /      C_110  [thesis prints C_011 TYPO]

double omega8 = CUM_OMEGA8;  // Can vary; default 1.0

// Velocity derivatives from CE analysis
double dywPdzv = -3.0 * omega * inv_rho * saved_C011;  // yz pair
double dxwPdzu = -3.0 * omega * inv_rho * saved_C101;  // xz pair
double dxvPdyu = -3.0 * omega * inv_rho * saved_C110;  // xy pair

// Eq. 3.86-3.88: General form with omega8
// C*_211 = -1/3(     /2-1)        B        (  _yw  +  _zv  ) + (1-        ) C_211
double eq_coeff = -1.0/3.0 * (omega * 0.5 - 1.0) * omega8 * coeff_B * rho;
wp_C211_star = eq_coeff * dywPdzv + (1.0 - omega8) * CUMcbb;
wp_C121_star = eq_coeff * dxwPdzu + (1.0 - omega8) * CUMbcb;
wp_C112_star = eq_coeff * dxvPdyu + (1.0 - omega8) * CUMbbc;
```

Listing 16: Generalized Eq. 3.86–3.88 with omega8 parameter

Verification when $\omega_8 = 1$:

$$\text{eq_coeff} \times \text{dywPdzv} = -\frac{1}{3}(\omega/2 - 1) \cdot 1 \cdot B \cdot \rho \cdot (-3\omega/\rho \cdot C_{011}) \tag{60}$$

$$= (\omega/2 - 1)\omega B C_{011} \tag{61}$$

$$(1 - 1) \cdot \text{CUMcbb} = 0 \tag{62}$$

$$\Rightarrow \text{wp_C211_star} = (\omega/2 - 1)\omega B C_{011} \quad \checkmark \tag{63}$$

This matches the current code when $\omega_8 = 1$.

7.6 Additional Requirements

To support this generalization, also **fix the back-conversion** (from Q2):

```
// WP mode: add cumulant-to-central moment back-conversion
m[I_cbb] = wp_C211_star + ((m[I_caa]+1.0/3.0)*m[I_abb] +
                            2.0*m[I_bba]*m[I_bab]) * inv_rho;
```

Listing 17: Back-conversion with general omega8

This requires computing `CUMcbb, CUMbcb, CUMbbc` in WP mode (currently only done in AO mode).

—

8 Question 8: Eq. 3.108–3.110 — Automatic Behavior

8.1 Thesis Reference

- **Eq. 3.108**: $\lim_{\nu\to 0} \omega_{3,4,5}(\omega) \to 0$
- **Eq. 3.109–3.110**: At low viscosity, $C^* \to C$ (cumulants conserved)

8.2 Code Location

Lines 78–87 (safety clamp) and lines 393–409 (limiter computation)

8.3 Verdict

CORRECT

Automatic analytical property — no separate code implementation needed

8.4 Detailed Analysis

8.4.1 Analytical Behavior

Eq. 3.108–3.110 describe the low-viscosity behavior of the parametrization:

1. **As** $\nu \to 0$:
$$\omega_1 \to 2 \quad \text{(from Eq. 3.66: } \omega_1 = 2/(6\nu+1)\text{)} \tag{64}$$

2. **Substituting into Eq. 3.103–3.105**:
$$\omega_3 = 2(\omega_1 - 1) \to 0 \tag{65}$$
$$\omega_4 = 3(\omega_1 - 1) \to 0 \tag{66}$$
$$\omega_5 = 3/2(\omega_1 - 1) \to 0 \tag{67}$$

3. **When** $\omega \to 0$:
$$C^* = (1-\omega)C + \omega C^{\text{eq}} \to C \quad \text{(full conservation)} \tag{68}$$

8.4.2 Automatic in Code

These are *properties of the mathematical formulas*, not algorithmic steps to implement:

```
// The formulas from Eq. 3.103-3.105 automatically encode
// the low-viscosity limit. No special code needed.
double omega3_base = 2.0 * (omega - 1.0);  // -> 0 as omega -> 2
double omega4_base = 3.0 * (omega - 1.0);  // -> 0 as omega -> 2
double omega5_base = 1.5 * (omega - 1.0);  // -> 0 as omega -> 2
```

Listing 18: Automatic low-viscosity behavior through parameterization

8.4.3 Lambda Limiter Counteracts Unconditional Conservation

The λ limiter (Eq. 3.111–3.114) prevents pathological behavior at low viscosity:

$$\tilde{\omega} = \omega_{\text{base}} + (1 - \omega_{\text{base}}) \frac{|C|}{\rho\lambda + |C|} \tag{69}$$

When cumulants are small ($|C| \ll \rho\lambda$):

$$\tilde{\omega} \approx \omega_{\text{base}} + (1 - \omega_{\text{base}}) \frac{|C|}{\rho\lambda} \tag{70}$$

When $\omega_{\text{base}} \to 0$ and $|C|$ becomes significant, $\tilde{\omega}$ is *pushed toward 1.0*, preventing unconditional conservation that would lead to instability.

8.4.4 Safety Clamp

Lines 78–87 provide additional protection:

```
static inline double _cum_wp_clamp_with_ao_fallback(double omega) {
    if (omega < 0.0 || omega > 2.0) {
        // omega out of bounds: fall back to AO neutral
        return 1.0;
    }
    return omega;
}
```

Listing 19: Safety clamp for omega values (cumulant_collision.h:78–87)

This handles singularities in Eq. 3.104 (when $\omega_1 \approx 14/9 \approx 1.556$, the denominator vanishes).

8.5 Verdict Confirmation

No separate implementation needed. The behavior is automatic from the parameterization, and the lambda limiter + safety clamp provide numerical stability. No changes required.

—

9 Question 9: ω_6, ω_7 General Form Claim

9.1 Thesis References

- **Eq. 3.83–3.85** (pages 45–46): Explicit structure with ω_6, ω_7
- **Page 47, §3.2.3**: Both WP and All-One variants set $\omega_6 = \omega_7 = 1$

9.2 Code Location

Lines 355–356 (parameter storage), lines 527–530 (equilibrium computation)

9.3 Verdict

WARNING/LIMITATION

Deviatoric/trace decomposition correct, but A,B equilibrium form is ONLY valid at $\omega_6 = \omega_7 = 1$

9.4 Detailed Analysis

9.4.1 Thesis Structure

The thesis Eq. 3.83–3.85 can be rewritten in deviatoric/trace form:

$$\kappa^*_{220} = \underbrace{[\text{velocity derivatives}]}_{\text{equilibrium}} \tag{71}$$

$$+ (1 - \omega_6)[\text{deviatoric modes}] \tag{72}$$

$$+ (1 - \omega_7)[\text{trace mode}] \tag{73}$$

The parameters ω_6, ω_7 appear explicitly.

9.4.2 Code Deviatoric/Trace Decomposition

The code correctly preserves this structure:

```
// Eq. 3.83-3.85: deviatoric and trace modes
// DEV1 = C_{220} - 2C_{202} + C_{022}  (deviatoric mode 1)
// TRACE = (C_{220} + C_{202} + C_{022}) / 3  (trace mode)

const double dev1 = /* ... */;
const double trace = /* ... */;

// A,B product form equilibrium
CUMcca_eq = (A * (Sxx*Syy + Dxy_sq) + B * (Sxx*Syy - Dxy_sq)) / rho;
```

Listing 20: 4th-order diagonal structure (cumulant_collision.h:527–530)

9.4.3 Equivalence Constraint

For the A,B form to match the thesis:

$$\omega_6 \cdot C^{\text{eq}}_{\text{A,B}} = \text{velocity_derivative_equilibrium}_{\text{thesis}} \tag{74}$$

This requires that when $\omega_6 \neq 1$, the equilibrium must be divided by ω_6. The code does *not* do this.

9.4.4 Parameter Status

Currently (lines 355–356):

```
const double omega6 = CUM_OMEGA6;  // Default 1.0 (hardcoded)
const double omega7 = CUM_OMEGA7;  // Default 1.0 (hardcoded)
```

Listing 21: Parameter values (cumulant_collision.h:355–356)

The parameters are defined but always set to 1.0, which maintains equivalence with the thesis.

9.4.5 Thesis Constraint

Page 47, §3.2.3 (Parameterized variant):

> "The remaining higher-order ($\bar{o} > 3$) relaxation rates are set to unity, i.e., $\omega_{C>5} = 1$"

This explicitly sets $\omega_6 = \omega_7 = \omega_8 = \omega_9 = \omega_{10} = 1$ for the Parameterized LBM variant used in the thesis.

9.5 Recommended Fix

Document the constraint clearly:

```
// CONSTRAINT: Thesis Eq. 3.83-3.85 with A,B equilibrium form is
//             ONLY valid when omega6 = omega7 = 1 (per thesis 3.2.3)
// For testing with general omega6, omega7 != 1, must:
// 1. Revert to explicit velocity derivative equilibrium form
// 2. Divide equilibrium by omega6 and omega7 as needed
const double omega6 = CUM_OMEGA6;  // MUST be 1.0 for current form
const double omega7 = CUM_OMEGA7;  // MUST be 1.0 for current form

// Add to variables.h:
#define CUM_OMEGA6 1.0  // Constraint: must be 1.0 for A,B equilibrium
#define CUM_OMEGA7 1.0  // Constraint: must be 1.0 for A,B equilibrium
```

Listing 22: Updated parameter documentation

—

10 Question 10: Generalize Eq. 3.86–3.88 with ω_8 Parameter

10.1 Thesis References

- **Eq. 3.86–3.88**: Off-diagonal 4th-order collision formulas
- **Eq. 3.89**: Velocity derivative approximation (with TYPO in subscripts)
- **Page 47**: $\omega_8 = 1$ for the Parameterized variant

10.2 Code Location

- Lines 264–269: Pre-collision 4th-order cumulants in AO mode
- Lines 566–571: Simplified form with $\omega_8 = 1$ (WP mode)
- Lines 613–615: Backward transformation in WP mode

10.3 Verdict

WARNING/LIMITATION

Currently simplified ($\omega_8 = 1$), but should be generalized to support arbitrary ω_8

10.4 Detailed Analysis

10.4.1 Step 1: Velocity Derivatives from Eq. 3.89 (CORRECTED)

The thesis Eq. 3.89 provides Chapman–Enskog approximations. **Note: subscripts are SWAPPED in the original thesis.**

Corrected relations:

$$\partial_y w + \partial_z v = -\frac{3\omega_1}{\rho} C_{011} \quad \text{(yz pair)} \tag{75}$$

$$\partial_x w + \partial_z u = -\frac{3\omega_1}{\rho} C_{101} \quad \text{(xz pair)} \tag{76}$$

$$\partial_x v + \partial_y u = -\frac{3\omega_1}{\rho} C_{110} \quad \text{(xy pair)} \tag{77}$$

Code implementation:

```
// Step 1: Compute velocity derivatives from 2nd-order cumulants
// Eq. 3.89 (CORRECTED for thesis typo):
//     _yw   +   _zv   = -3     /      C_011  [thesis prints C_110
    TYPO]
```

```
//      _xw    +    _zu    = -3      /       C_101   [correct]
//      _xv    +    _yu    = -3      /       C_110   [thesis prints C_011
    TYPO]

double dywPdzv = -3.0 * omega * inv_rho * saved_C011;  // yz pair
double dxwPdzu = -3.0 * omega * inv_rho * saved_C101;  // xz pair
double dxvPdyu = -3.0 * omega * inv_rho * saved_C110;  // xy pair
```

Listing 23: Velocity derivatives from Eq. 3.89 (corrected)

10.4.2 Step 2: General Eq. 3.86–3.88 with ω_8

General collision formula (keeping ω_8 as parameter):

$$\kappa^*_{211} = -\frac{1}{3}(\omega_1/2 - 1)\omega_8 B\rho(\partial_y w + \partial_z v) + (1 - \omega_8)\kappa_{211} \tag{78}$$

$$\kappa^*_{121} = -\frac{1}{3}(\omega_1/2 - 1)\omega_8 B\rho(\partial_x w + \partial_z u) + (1 - \omega_8)\kappa_{121} \tag{79}$$

$$\kappa^*_{112} = -\frac{1}{3}(\omega_1/2 - 1)\omega_8 B\rho(\partial_x v + \partial_y u) + (1 - \omega_8)\kappa_{112} \tag{80}$$

Code implementation (generalized):

```
// Step 2: Compute equilibrium coefficient
// Eq. 3.86-3.88: C* _       = -1/3(     /2-1)        B        (    +
   c terms) + (1-     )     C_

const double omega8 = CUM_OMEGA8;  // Default 1.0, can be varied
const double eq_coeff = -1.0/3.0 * (omega * 0.5 - 1.0) * omega8 *
   coeff_B * rho;

// Step 3: Compute WP-corrected off-diagonal 4th-order cumulants
// (Note: requires CUMcbb, CUMbcb, CUMbbc from pre-collision state)
double wp_C211_star = eq_coeff * dywPdzv + (1.0 - omega8) * CUMcbb;
double wp_C121_star = eq_coeff * dxwPdzu + (1.0 - omega8) * CUMbcb;
double wp_C112_star = eq_coeff * dxvPdyu + (1.0 - omega8) * CUMbbc;
```

Listing 24: Generalized Eq. 3.86-3.88 with omega8

10.4.3 Step 3: Back-Conversion with Product Terms (from Q2)

Critical fix: add cumulant-to-central moment back-conversion

```
// Step 4: Back-convert cumulant to central moment
// Eq. J.16 (Geier 2015): must add products of lower-order cumulants
m[I_cbb] = wp_C211_star + ((m[I_caa]+1.0/3.0)*m[I_abb] +
                            2.0*m[I_bba]*m[I_bab]) * inv_rho;
m[I_bcb] = wp_C121_star + ((m[I_aca]+1.0/3.0)*m[I_bab] +
                            2.0*m[I_bba]*m[I_abb]) * inv_rho;
```

```
m[I_bbc] = wp_C112_star + ((m[I_aac]+1.0/3.0)*m[I_bba] +
                            2.0*m[I_bab]*m[I_abb]) * inv_rho;
```

Listing 25: WP mode back-conversion with product terms

10.4.4 Verification: Simplified Case $\omega_8 = 1$

Substituting $\omega_8 = 1$:

$$\text{eq_coeff} = -\frac{1}{3}(\omega/2 - 1) \cdot 1 \cdot B \cdot \rho \tag{81}$$

$$\text{wp_C211_star} = \text{eq_coeff} \cdot \text{dywPdzv} + 0 \tag{82}$$

$$= -\frac{1}{3}(\omega/2 - 1)B\rho \cdot (-3\omega/\rho \cdot C_{011}) \tag{83}$$

$$= (\omega/2 - 1)\omega B C_{011} \tag{84}$$

This matches the current code (lines 566–571) when $\omega_8 = 1$.

10.4.5 Implementation Checklist

To fully implement Q10:

1. **Add ω_8 parameter to variables.h**:

```
#define CUM_OMEGA8 1.0  // 4th-order off-diagonal relaxation
```

Listing 26: Add to variables.h

2. **Compute pre-collision 4th-order cumulants in WP mode**:

```
// Pre-collision 4th-order off-diagonal cumulants (needed for
   Eq. 3.86-3.88)
double CUMcbb = m[I_cbb];  // pre-collision value
double CUMbcb = m[I_bcb];  // pre-collision value
double CUMbbc = m[I_bbc];  // pre-collision value
```

Listing 27: Add to WP mode (before line 566)

3. **Update Eq. 3.86–3.88 computation**: Replace lines 566–571 with the generalized form above.

4. **Update back-conversion**: Replace lines 613–615 with the corrected form that includes product terms (from Q2).

5. **Document the Eq. 3.89 typo correction**: Add comment clarifying that subscripts are swapped in the thesis.

10.5 Recommended Complete Code Section

```
// ==========================================================================
// Eq. 3.86-3.88 (with omega8 generalization)
// ==========================================================================
// Note: Eq. 3.89 velocity derivatives are independent of omega8
//       (they come from CE analysis of 2nd-order collision)

const double omega8 = CUM_OMEGA8;  // Default 1.0

// Step 1: Velocity derivatives from Eq. 3.89 (CORRECTED subscripts)
//     _yw   +   _zv   = -3     /   C_011     [thesis prints C_110 TYPO]
//     _xw   +   _zu   = -3     /   C_101
//     _xv   +   _yu   = -3     /   C_110     [thesis prints C_011 TYPO]
double dywPdzv = -3.0 * omega * inv_rho * saved_C011;
double dxwPdzu = -3.0 * omega * inv_rho * saved_C101;
double dxvPdyu = -3.0 * omega * inv_rho * saved_C110;

// Step 2: Pre-collision 4th-order cumulants
double CUMcbb = m[I_cbb];
double CUMbcb = m[I_bcb];
double CUMbbc = m[I_bbc];

// Step 3: Equilibrium coefficient
// Eq. 3.86: C*_211 = -1/3(     /2-1)        B        (  _yw  +  _zv  ) + (1-     )  C_211
double eq_coeff = -1.0/3.0 * (omega * 0.5 - 1.0) * omega8 * coeff_B * rho;

// Step 4: WP-corrected off-diagonal 4th-order cumulants
double wp_C211_star = eq_coeff * dywPdzv + (1.0 - omega8) * CUMcbb;
double wp_C121_star = eq_coeff * dxwPdzu + (1.0 - omega8) * CUMbcb;
double wp_C112_star = eq_coeff * dxvPdyu + (1.0 - omega8) * CUMbbc;

// Step 5: Back-conversion (cumulant     central moment, with products)
m[I_cbb] = wp_C211_star + ((m[I_caa]+1.0/3.0)*m[I_abb] +
                           2.0*m[I_bba]*m[I_bab]) * inv_rho;
m[I_bcb] = wp_C121_star + ((m[I_aca]+1.0/3.0)*m[I_bab] +
                           2.0*m[I_bba]*m[I_abb]) * inv_rho;
m[I_bbc] = wp_C112_star + ((m[I_aac]+1.0/3.0)*m[I_bba] +
                           2.0*m[I_bab]*m[I_abb]) * inv_rho;
```

Listing 28: Complete generalized Q10 implementation

—

Summary: All Questions at a Glance

Q	Verdict	Issue	Impact/Recommendation
		CRITICAL BUG	
Q2		WP mode missing back-conversion products	**FIX IMMEDIATELY**: Add product terms to WP mode
		WARNINGS / PARAMETERIZATION CONSTRAINTS	
Q1		A,B form only valid at $\omega_6 = \omega_7 = 1$	Update documentation; keep form unless testing general ω_6, ω_7
Q6		A,B equilibrium only valid at $\omega_6 = \omega_7 = 1$	Same as Q1; document constraint
Q7		$\omega_8 = 1$ assumed; could be generalized	Optional enhancement; currently correct
Q9		ω_6, ω_7 form claim needs clarification	Document that A,B form only valid at $\omega_6 = \omega_7 = 1$
Q10		$\omega_8 = 1$ hardcoded; could be parameter	Enhancement + Q2 fix needed for full generalization
		CORRECT IMPLEMENTATIONS	
Q3		Half-force correction location	No changes needed; correctly implements Strang splitting
Q4		Velocity derivative equations 3.73–3.75	No changes needed; efficient incremental form correct
Q5		Tilde-omega limiter function	No changes needed; all 7 pairs correctly matched
Q8		Low-viscosity behavior (Eq. 3.108–3.110)	No changes needed; automatic from parameterization

Table 3: Summary of all 10 questions: verdict and recommended actions

—

Conclusion

This analysis identifies **one critical bug** (Q2: missing back-conversion in WP mode) that must be fixed immediately, as it directly affects the accuracy of off-diagonal 4th-order moment evolution near walls.

Additionally, four warning-level issues (Q1, Q6, Q7, Q9, Q10) require documentation updates and/or optional enhancements for future parameterization flexibility.

The remaining implementations (Q3–Q5, Q8) are correct and well-designed.

References

[1] Gehrke, *Lattice Boltzmann Methods for Fluid Dynamics: Theory, Cumulant Collision Operator, and GPU Implementation*, PhD Thesis, Technische Universität München, 2024.

[2] Geier, M., Schönherr, M., Pasquali, A., & Krafczyk, M., "The Chapman–Enskog expansion of the lattice Boltzmann method: The off-equilibrium distribution in the incompressible limit", *Journal of Fluid Mechanics*, vol. 777, pp. 564–611, 2015.

[3] Gehrke, M., & Ritter, A., "A simplified formulation of the fourth-order diagonal equilibrium", unpublished technical note, 2022.